

Literature-Implied Answer to Open Problem 5.2 of arXiv:1512.00419

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Status. `literature_implied_answer`, scoped to Open Problem 5.2. This packet records a later-literature answer to the finite-game infinite-entanglement question in Palazuelos–Vidick’s survey [1]. It is not a new proof produced by the run.

Source question. Palazuelos and Vidick, Section 5.4.1, page 42, explain that the then-known examples either used quantum questions or infinitely many possible answers, and therefore did not yet give a “standard” multiplayer game whose entangled value is not attained in finite dimension. They ask:

Does there exist a multiplayer game G such that the entangled value $\omega^(G)$ is only attained in the limit of infinite-dimensional entanglement?*

Due to their use respectively of quantum questions or an infinite number of possible answers these examples do not quite provide a construction of a “standard” multiplayer game for which the entangled value is not attained by any finite-dimensional strategy, and at the time of writing no such example has been found. Perhaps the best candidate is the I_{3322} Bell inequality for which suggestive numeric investigations have been performed [PVI0].

Open problem 5.2. *Does there exist a multiplayer game G such that the entangled value $\omega^*(G)$ is only attained in the limit of infinite-dimensional entanglement?*

It is interesting to investigate the rate at which the entangled value can be approached, as a function of the dimension. The games described above suggest some lower bounds, but no upper bound is known.

Open problem 5.3. *Given $\epsilon > 0$ and $n \in \mathbb{N}$ provide an explicit^[24] bound on $d = d(\epsilon, n)$ such that for any game G with at most n questions and answers per player there exists a d -dimensional strategy with success probability at least $\omega^*(G) - \epsilon$.*

We believe answering these two natural questions poses a significant challenge; indeed in spite of increased interest over the years little concrete progress has been made.

Identification. The answer is yes, by later finite nonlocal-game constructions. Slofstra [2] constructs a finite linear-system nonlocal game that can be played perfectly using a limit of finite-dimensional quantum strategies, but cannot be played perfectly on any finite-dimensional Hilbert space, or even with any tensor-product strategy. In the paper’s notation, Theorem 1.1 gives a finite nonlocal game with a perfect strategy in C_{qa} and no perfect strategy in C_{qs} .

Ji, Leung, and Vidick [3] give an especially direct finite three-player example. Their abstract states that the game has finitely many classical questions and answers and that optimal success probability 1 can only be achieved in the limit of arbitrarily high-dimensional entangled states. Their Theorem 1 gives a concrete three-player game: each player has 12 possible questions, the

answer alphabets are finite, and achieving success probability $1 - \varepsilon$ forces the local dimension for two players to grow at least as $2^{\Omega(\varepsilon^{-c})}$ for some constant $c > 0$.

Why this is an implied literature answer. The checked sources do not appear to explicitly say that they are resolving Palazuelos–Vidick Open Problem 5.2. The implication is nevertheless direct: both supporting papers give ordinary finite nonlocal games, while Open Problem 5.2 asks only for the existence of such a multiplayer game whose value is attained only in the infinite-dimensional limit.

Scope limitations. This packet clears only Open Problem 5.2 from [1]. It does not answer Open Problem 5.3 on explicit uniform bounds $d(\varepsilon, n)$ for approximating $\omega^*(G)$ for arbitrary games with at most n questions and answers per player. The Ji–Leung–Vidick theorem gives quantitative dimension growth for its particular game, not the uniform arbitrary-game upper bound asked for in Open Problem 5.3.

Search evidence. The run’s lightweight indexes were searched for 1512.00419, the source title, and core phrases including **infinite-dimensional entanglement**, **linear system non-local game**, and **coherent state embezzlement**. No existing packet for arXiv:1512.00419 was found. The answer evidence comes from arXiv pages and local PDFs for [2, 3].

References

- [1] Carlos Palazuelos and Thomas Vidick. *Survey on Nonlocal Games and Operator Space Theory*. arXiv:1512.00419, 2015. <https://arxiv.org/abs/1512.00419>.
- [2] William Slofstra. *The set of quantum correlations is not closed*. arXiv:1703.08618, 2017. <https://arxiv.org/abs/1703.08618>.
- [3] Zhengfeng Ji, Debbie Leung, and Thomas Vidick. *A three-player coherent state embezzlement game*. arXiv:1802.04926, 2018; Quantum 4, 349, 2020. <https://arxiv.org/abs/1802.04926>.